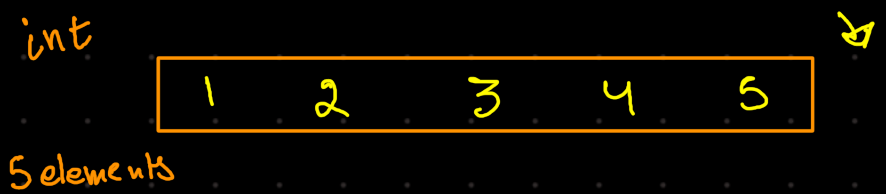
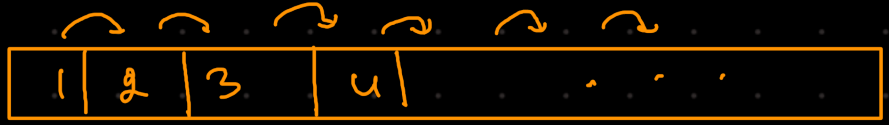
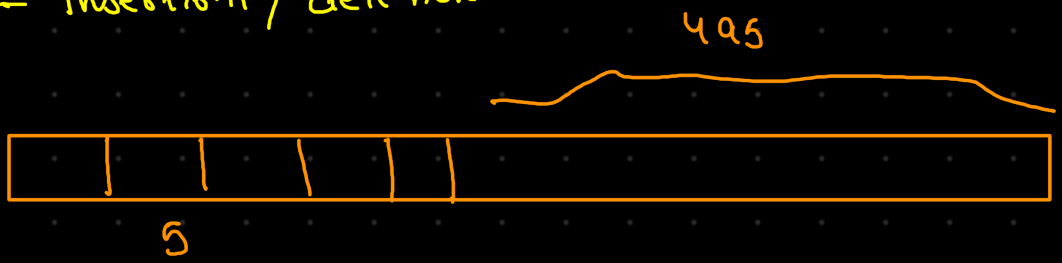
Linked List

First let us understand why linked list?





- insertion / deletion

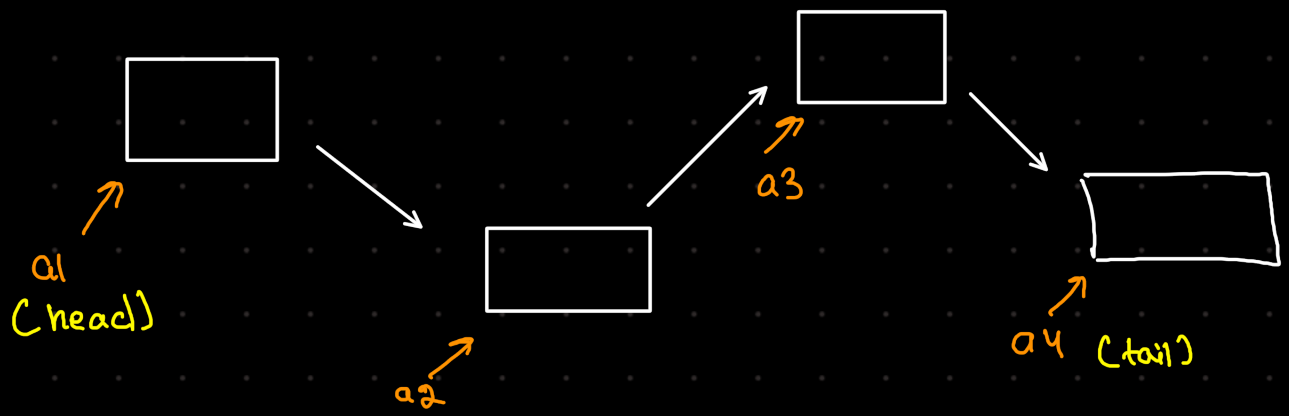


- memory wasted a lot many times

Linked list created to tackle this issue

⇒ add element as per requirement

Random allocation in space. (non-continous)

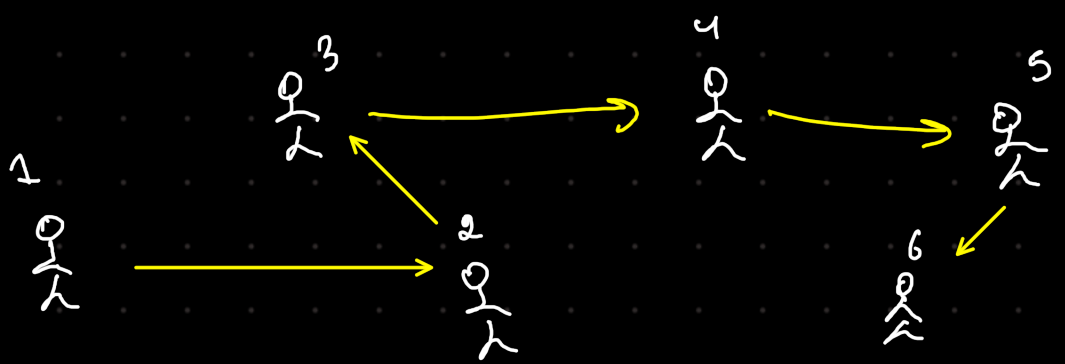


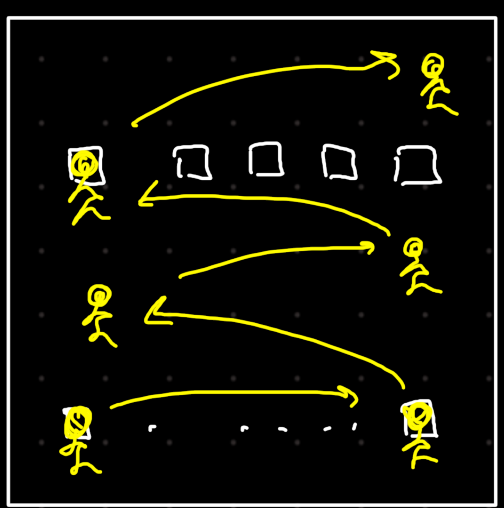
1. Create a list/array? ✗✗



were avoiding ✗ array.

2. Will ~~&~~ just store address of 1st and access other elements via that





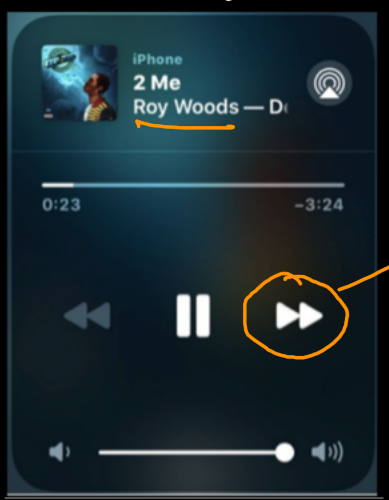
data	next

Classes
↓↓
user defined datatype

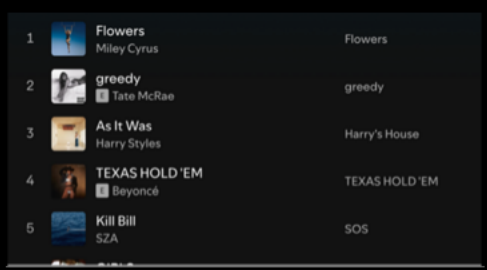
Node

Few great examples. :

1. Music Playlist



Save Address/reference



Song

2. Treasure hunt



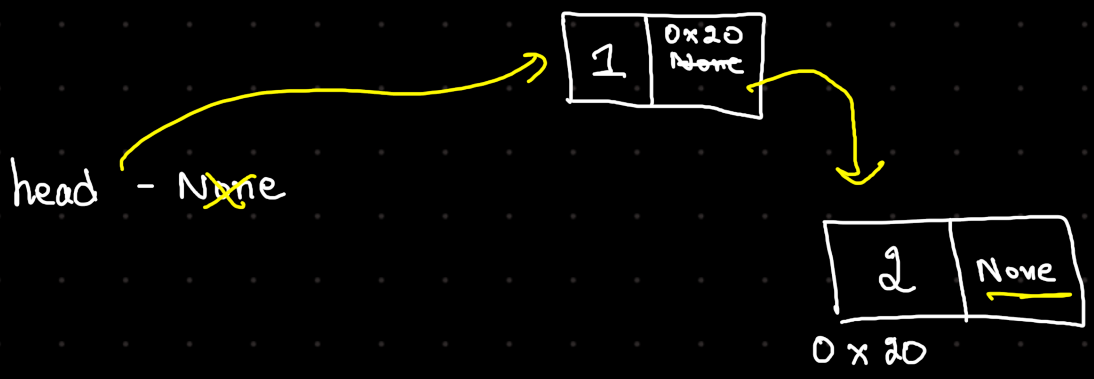
Linked List : Define and Print



Print

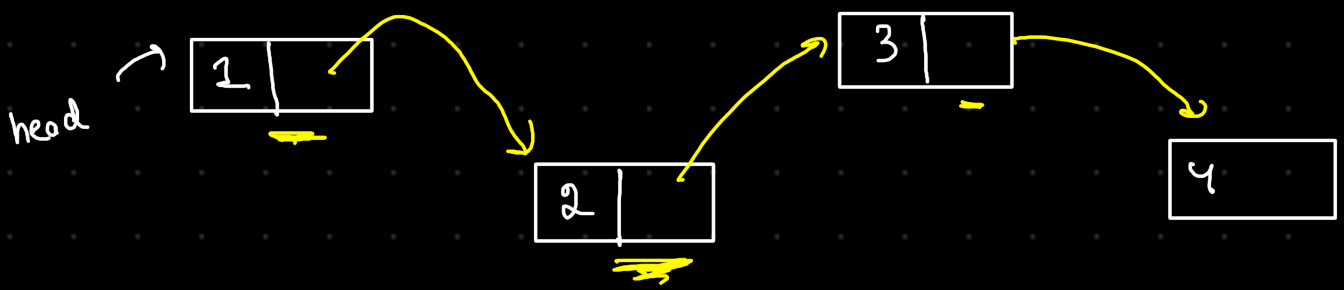
- Make sure to create a temp variable
- Iteratively and recursively.

Take Input



1

Take Input right




```python
def take_input_Right():
    head = None
    data = int(input("Enter the Data"))

    while(data != -1):
        newNode = Node(data)
        if(head == None):
            head = newNode
        else:
            temp = head
            while(temp!=None and temp.next !=None):
                temp = temp.next

            temp.next = newNode

        data = int(input("Enter the Data"))

    return head
```

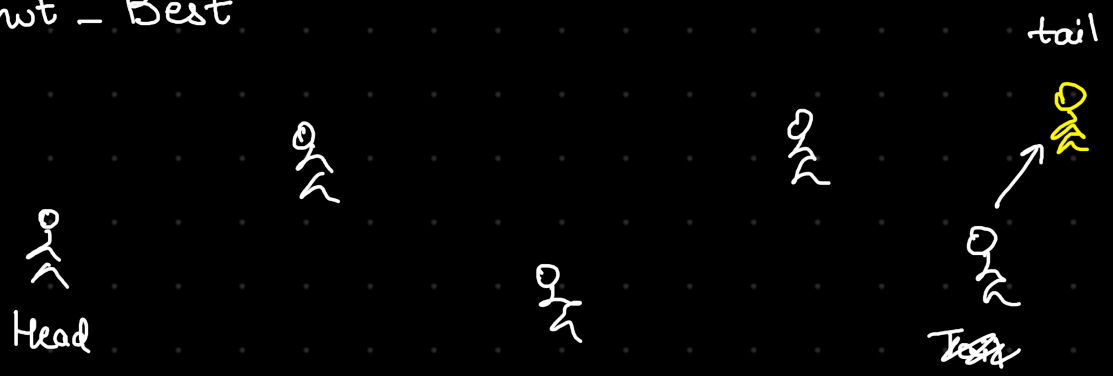
$1 \rightarrow 0$
$2 \rightarrow 1$
$3 \rightarrow 2$
$4 \rightarrow 3$
$\vdots$
$n \rightarrow n-1$

$O(n^2)$

$\Sigma\ 1+2+3+4 \cdots n$

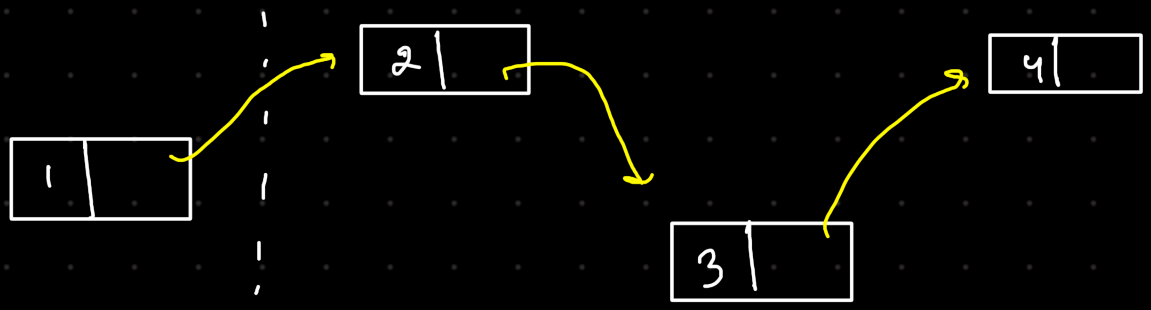
$= \frac{n(n+1)}{2} = n^2$

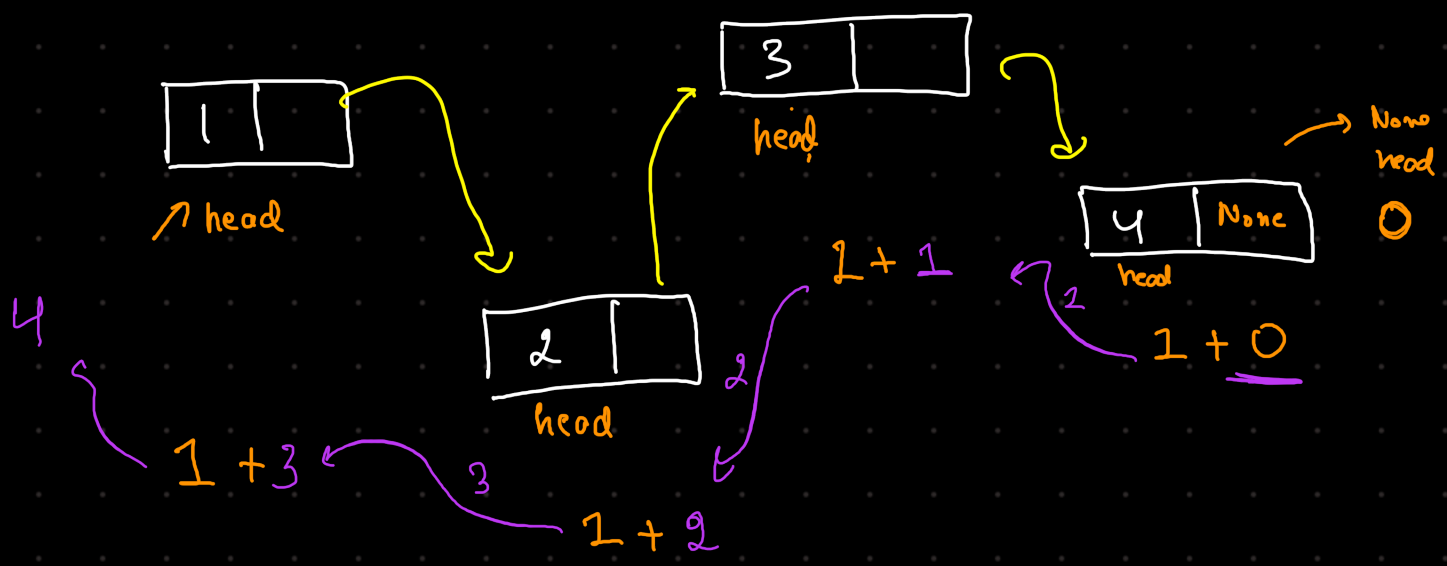
take input - Best



We will be keeping a head and tail this time.

Length of a Linked List





Linked List operations

1. Insert → head , tail (append) , in between
2. Delete → head , tail , index , value (↳ remove)
3. Search → value , index
4. Traverse → Print
5. Take Input

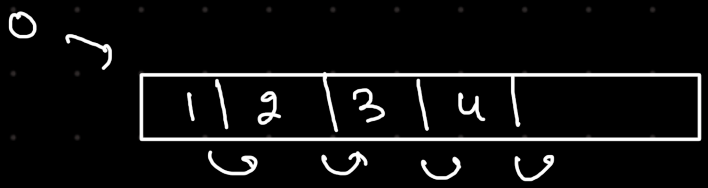
Insert

1. Head

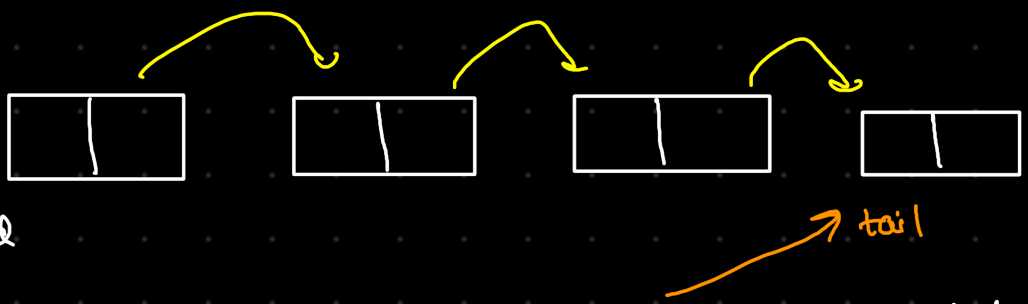




vs array



Insert at tail



head

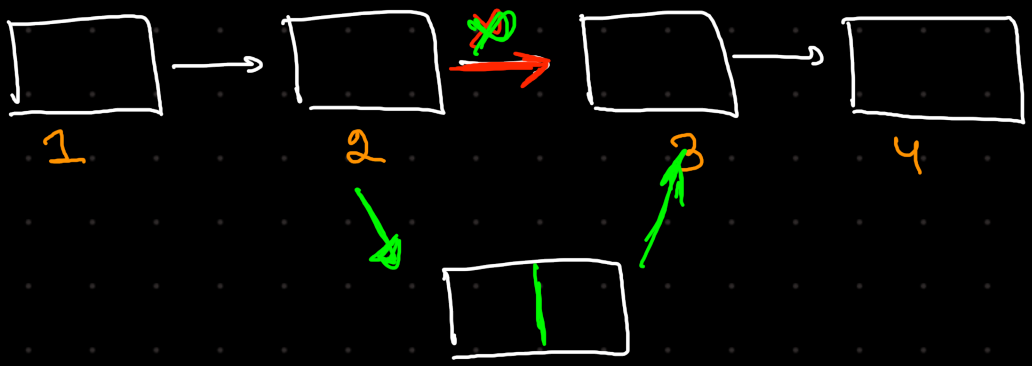
tail

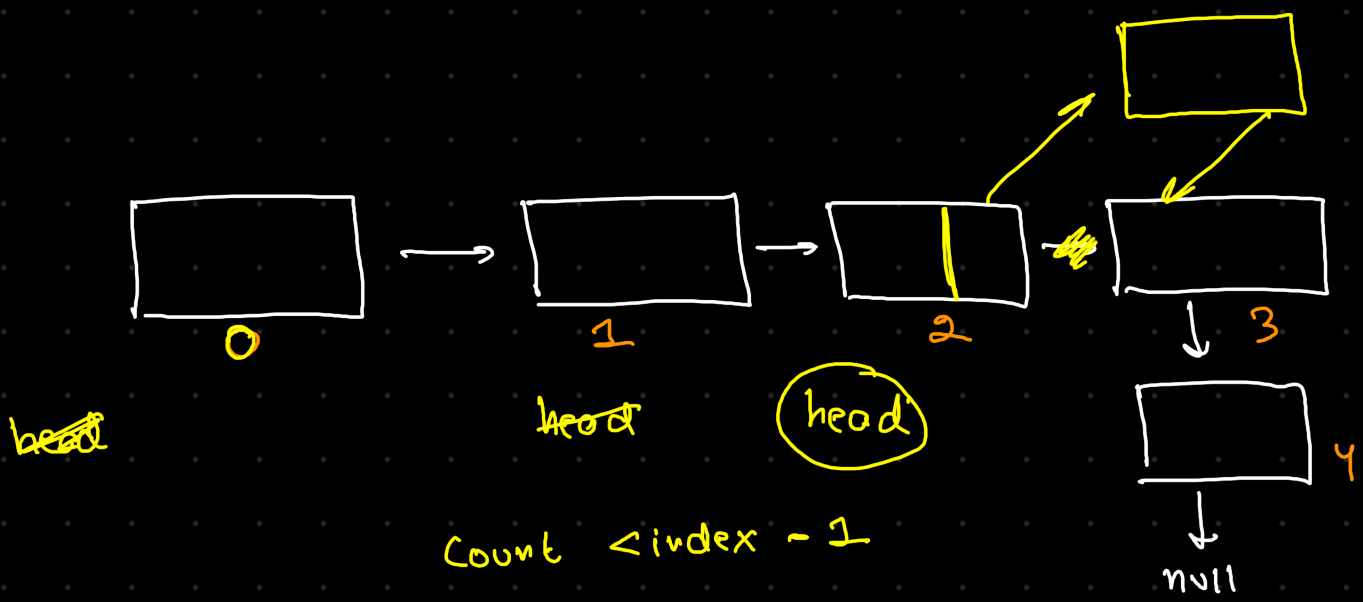
```
if (head.next != NULL)
```

vs list / array



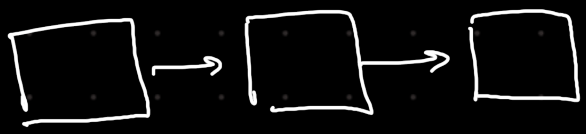
Insert at an Index





Delete.

1. Head



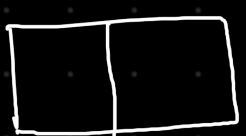


just return head.next

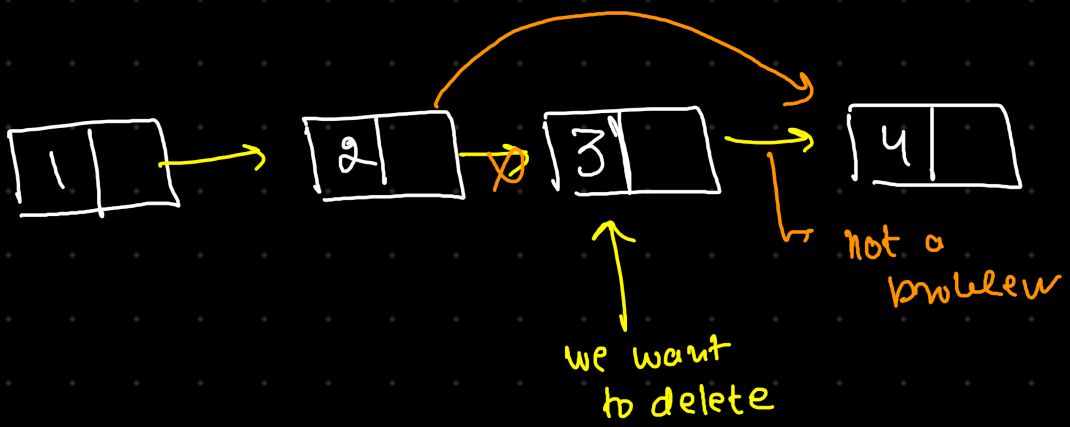
if head is not None

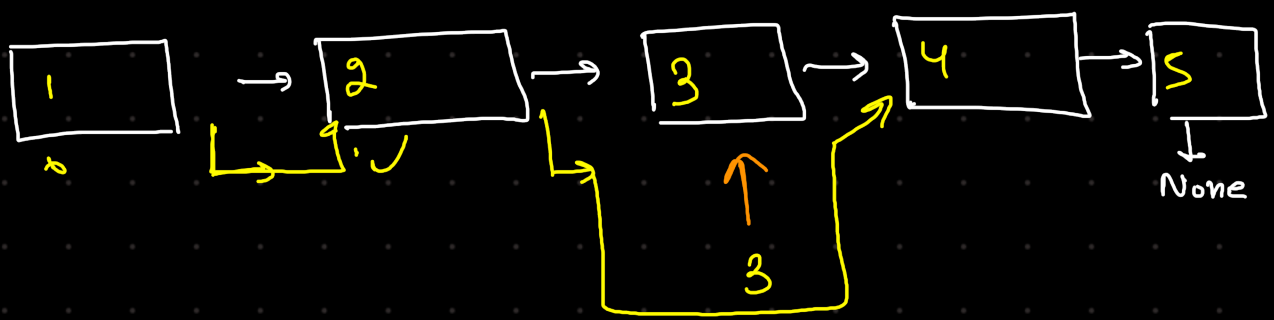
2. Tail





3. Delete at value (remove)





4] Delete by index

